



SENTRY

USER MANUAL



AI Powered Behavioral Analysis, Drive Health Prediction and Remote Monitoring for PC Users

Sentry is a Windows desktop monitoring system. It observes user and system behavior, compares new activity with learned baselines, records anomalies, and can send alerts through Telegram.

SYSTEM REQUIREMENTS

- Windows 10 or 11
- Python 3.10 or newer
- Internet access if Telegram alerts are required
- A Telegram bot token and chat ID for Telegram features

1. Before You Start

Use Sentry only on a computer and account where monitoring is authorized. The system records keyboard and mouse behavior, network usage, drive health, active window activity, and event logs.

You need:

- Windows 10 or 11
- Python 3.10 or newer
- Internet access if Telegram alerts are required
- A Telegram bot token and chat ID for Telegram features

Install the application dependencies:

```
pip install python-dotenv psutil pyautogui pygetwindow pyperclip requests plyer winshell  
pyttsx3 screen-brightness-control pygame pyTelegramBotAPI pandas numpy scikit-learn joblib  
pynput matplotlib PySide6
```

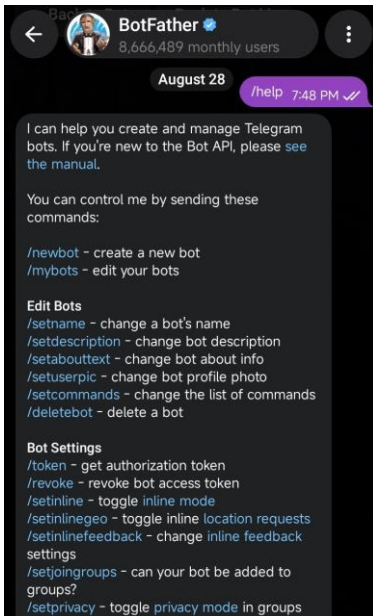
2. Get the Environment Variables

Sentry reads its private settings from a file named `.env`. Create that file beside the Sentry application files and add:

```
TELEGRAM_BOT_TOKEN=your_bot_token  
TELEGRAM_CHAT_ID=your_chat_id  
DASHBOARD_PASSWORD=your_dashboard_password
```

Telegram bot token

1. Open Telegram and search for `@BotFather`.
2. Send `/newbot` and follow the prompts for a bot name and username.
3. BotFather replies with an HTTP API token. Copy paste the whole token after `TELEGRAM_BOT_TOKEN=`.

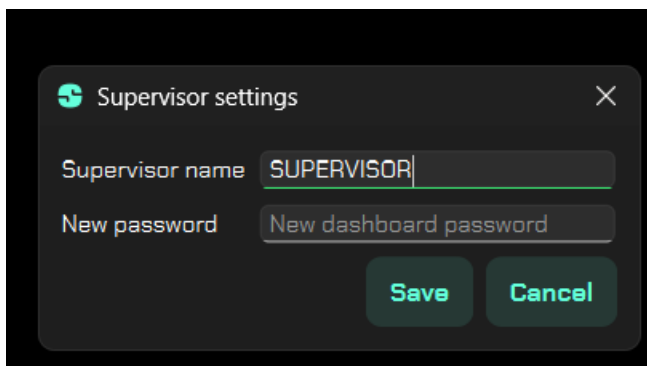


Telegram chat ID

1. Open the new bot and send it `/start`.
2. In a private browser window, open this address, replacing `YOUR_TOKEN` with the token from BotFather:
`https://api.telegram.org/botYOUR_TOKEN/getUpdates`
3. Find `message`, then `chat`, then `id` in the response. Copy that number after `TELEGRAM_CHAT_ID=`. Do not share the URL or response because it contains the bot token.

`DASHBOARD_PASSWORD` is the password used on the web login and for Telegram bot authentication. Choose a strong password. If it is omitted, the first launch asks `Set dashboard password:` in PowerShell. Type the password and press Enter; PowerShell does not display the characters while you enter them. Sentry stores only a derived hash in `settings.json`, not the plain password.

The password is created only when Sentry has no dashboard password yet. To change it later, right-click the Sentry tray icon and choose **Show Status**, then select the button at the top of the status panel. Enter the supervisor name and the new dashboard password, then confirm. The new password is immediately used for both the dashboard and Telegram authentication. Changing `DASHBOARD_PASSWORD` in `.env` does not replace an already stored password.



Telegram is optional for local monitoring, but the bot controller and notifications cannot work without the token and chat ID. Keep `.env` private.

The optional `SENTRY_DASHBOARD_URL` value can override the address opened by the tray. Normally leave it unset so Sentry uses the local-network address it prints when it starts. Do not add `SENTRY_DASHBOARD_TOKEN`; Sentry creates its internal dashboard token automatically.

3. Start Sentry and Open the Dashboard

Start the main application:

```
python .\main.pyw
```

When startup finishes, read the line beginning `Local network dashboard:` in the PowerShell window. Open the complete URL shown. example:

```
http://192.168.1.25:8765/login
```

Use the same local-network URL from another device on the same network. Always use the complete local-network address printed by Sentry. A different port can be selected by adding it to the command, for example `python .\main.pyw 8766`; use the printed address with that port.

Sign in with the dashboard password. The dashboard refreshes its status automatically while Sentry is running.

The first launch starts activity logging and the Telegram controllers. The four detection modules are not automatically enabled. In the dashboard, use **ENABLE ALL**, or enable these modules individually:

- Keystroke Dynamics
- Mouse Dynamics
- Network Usage
- Drive Health

To finish a session, use **STOP SENTRY** in the dashboard, confirm the action.

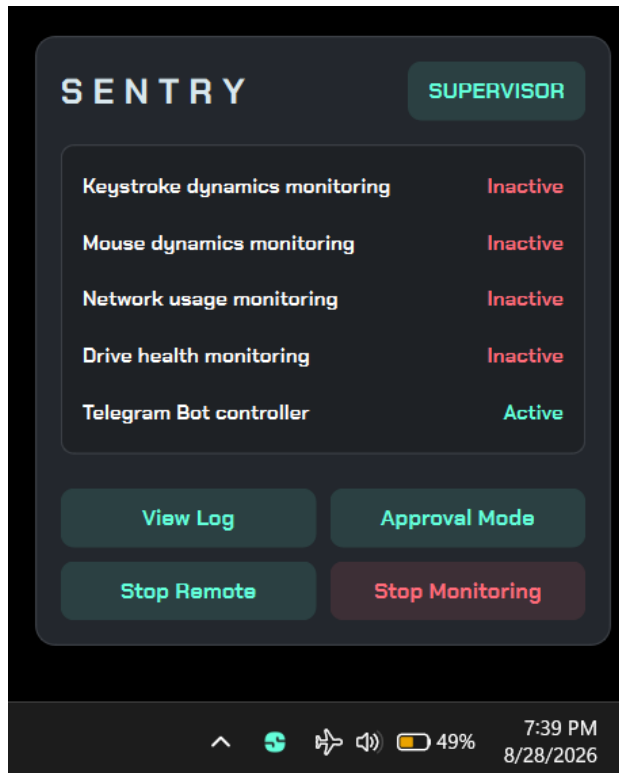
4. Use the System Tray

Sentry places an icon in the Windows notification area. If it is hidden, open the notification-area overflow menu. A single click or double-click opens the status panel, which refreshes every few seconds.

The status panel shows the state of keystroke, mouse, network, drive, and Telegram components. Its controls are:

- **View Log** opens the newest Sentry activity log. If no log is available, it opens the dashboard.
- **Allow All Mode** changes whether Telegram commands are accepted without the normal per-command restriction. Enable it only when you understand the risk.
- **Stop Remote** stops the Telegram remote controller but leaves other monitoring components running.
- **Stop Monitoring** shuts down Sentry and its components.
- **SUPERVISOR** lets you change the supervisor name shown by the tray.

Right-click the tray icon for **View Log**, **Show Status**, **Hide Status**, and **Stop Monitoring**. The tray icon is a quick status and shutdown tool; use the dashboard for detector controls, training, history, and reports.



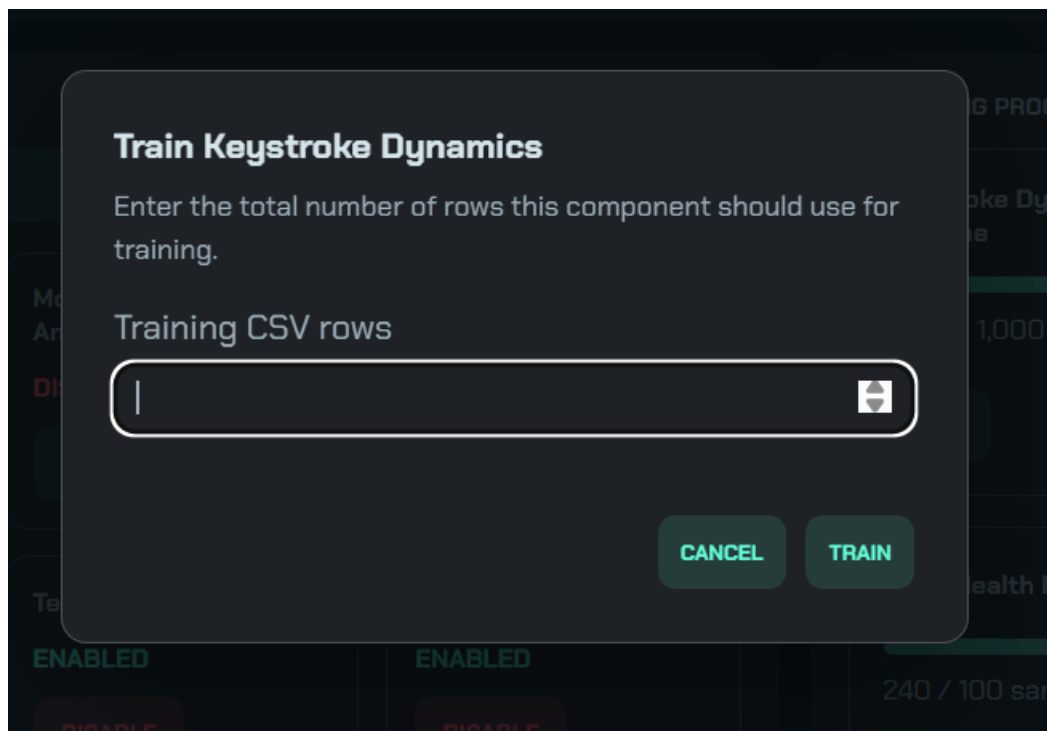
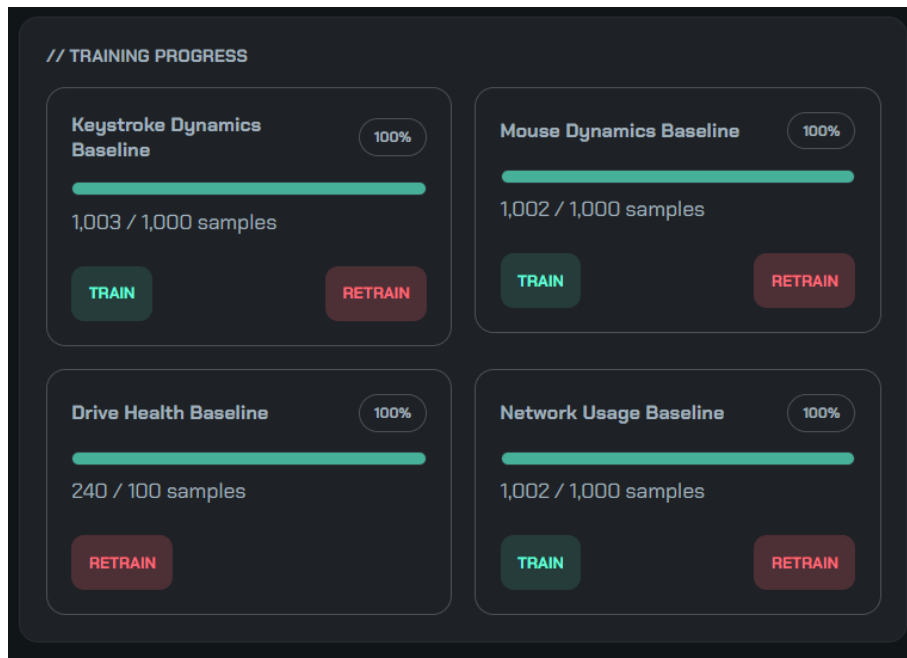
5. Train the Detection Baselines

Training teaches each detector what normal activity looks like for the monitored user or computer. Start training during representative normal use. Avoid training while another person other than the designated person is using the computer.

1. Start Sentry and sign in.
2. On **OVERVIEW**, find **Training Progress**.
3. Select **TRAIN** for Keystroke Dynamics, Mouse Dynamics, or Network Usage.
4. Enter the total number of CSV rows wanted.
5. Select **TRAIN** in the dialog. The detector may restart and its button will show `TRAINING...`
6. Continue representative normal activity until the progress bar reaches 100% and the status returns to ready.
7. Start the detector with its component **ENABLE** button or use **ENABLE ALL**.

The normal starting targets are 1,000 samples for keystroke, mouse, and network detectors, and 100 samples for drive health. The dashboard accepts a target from 100 to 1,000,000 rows. Use at least the normal target where possible; too few representative rows can make the model detections inaccurate. The training card shows completed rows, target rows, percentage, and the current state. Keystroke samples are collected from short typing windows (10s), mouse samples require mouse movement, network samples come from network activity, and drive samples come from SMART drive telemetry. Leave the relevant component running during collection.

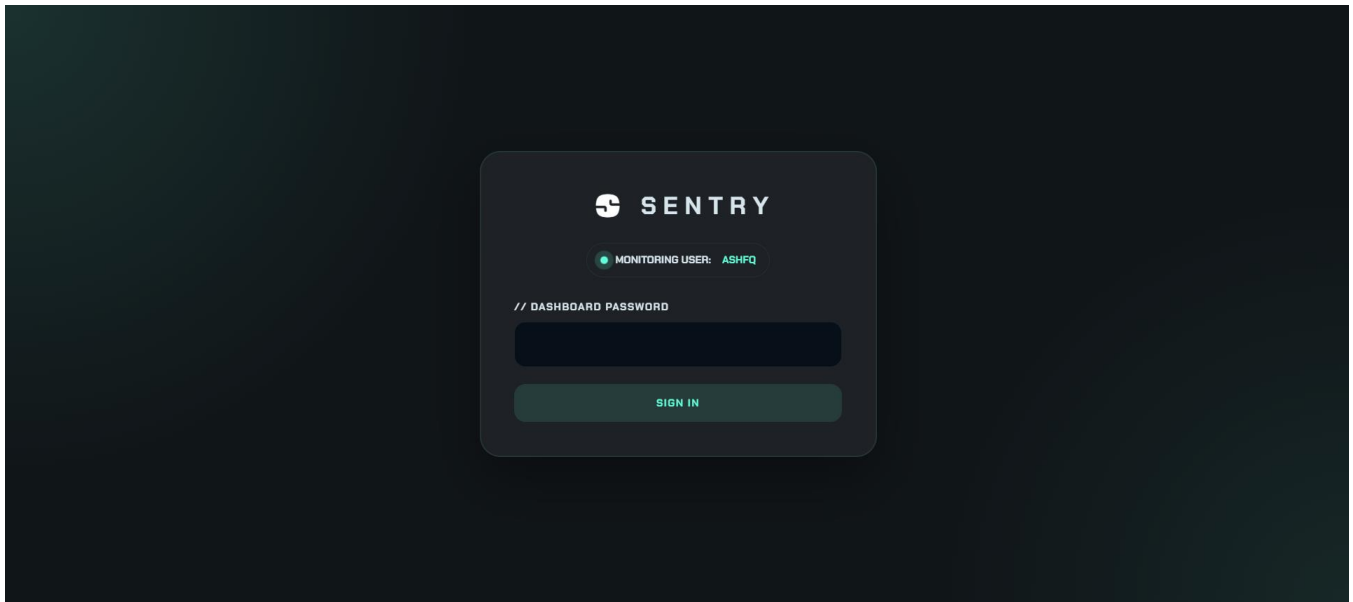
Use **RETRAIN** when the baseline no longer represents normal behavior, such as after a major change in the user's habits or computer environment. Enter the total replacement size, confirm, and wait for the progress to finish. Retraining removes the selected training CSV and model artifacts before rebuilding them, so existing baseline data for that detector is replaced. Do not retrain while unusual activity is occurring.



6. Use the Dashboard

Sign-in and status

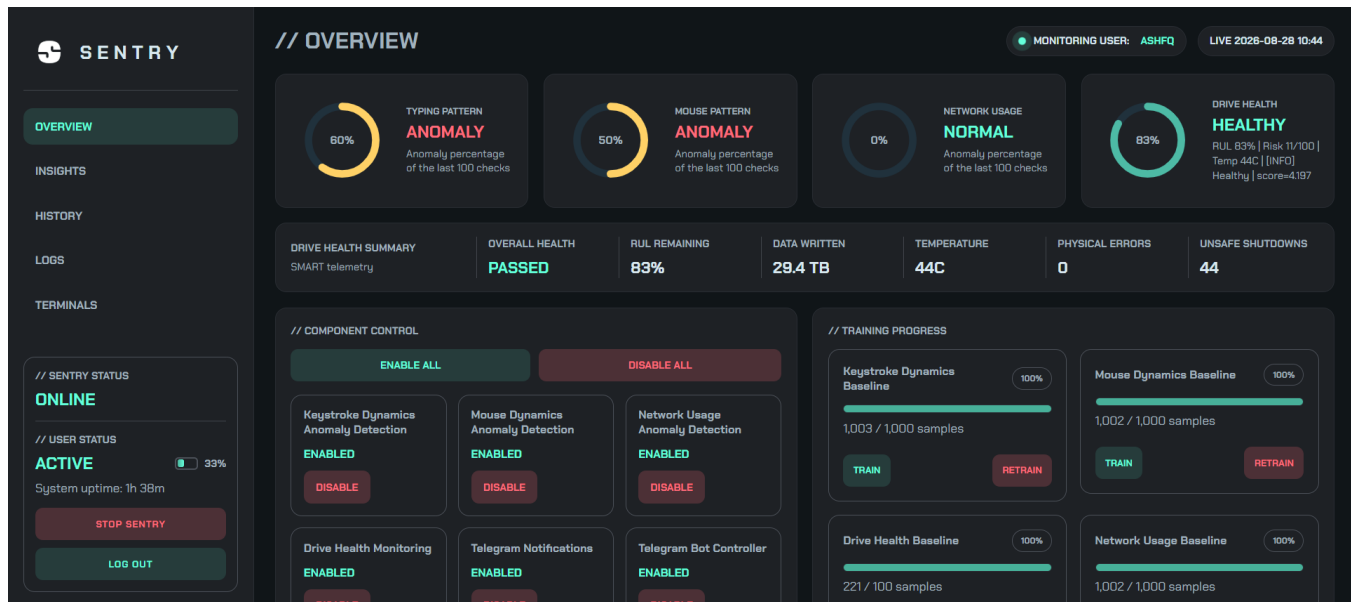
The login screen identifies the monitored Windows user and accepts the dashboard password. Repeated failed passwords temporarily lock access. After sign-in, the sidebar shows overall Sentry status, user status, battery information when available, and system uptime. **LOG OUT** ends the current dashboard session; it does not stop monitoring.



Overview

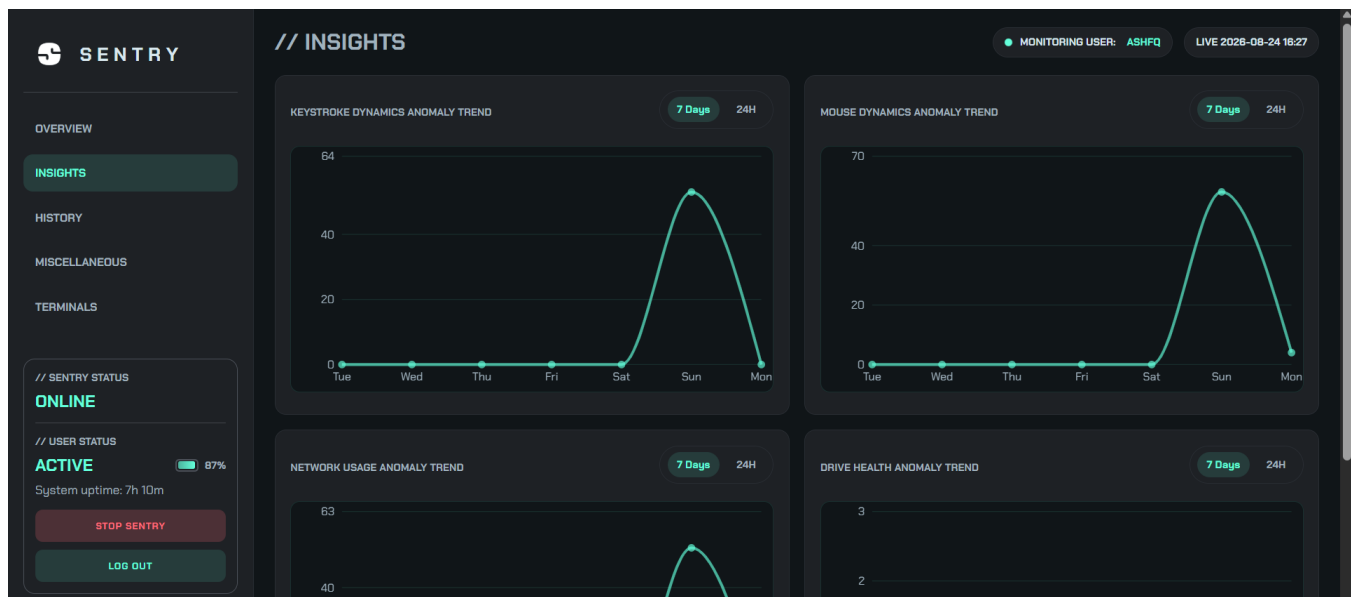
The **OVERVIEW** page contains four live meters for typing pattern, mouse pattern, network usage, and drive health. The drive section also shows SMART details such as overall health, Remaining Useful Life (RUL), data written, temperature, physical errors, and unsafe shutdowns. Use **Component Control** to enable or disable individual modules, or use **ENABLE ALL** and **DISABLE ALL**.

The same page contains the training cards. Each card shows baseline progress and offers **TRAIN** or **RETRAIN** where supported. A module showing `training` or `retraining` is still collecting or rebuilding data.



Insights

The **INSIGHTS** page shows anomaly trends for keystroke, mouse, network, and drive activity. Each chart can show the last **7 Days** or **24H**. Empty charts mean that no data is available for the selected period; they do not necessarily mean that the detector is disabled.



History

The **HISTORY** page shows recent alerts. Use **GENERATE REPORT** to download a PDF summary containing risk, detector distribution, trends, and recent events. **DELETE DETECTION HISTORY** permanently clears detection CSV data after confirmation; it does not retrain the models.

Logs

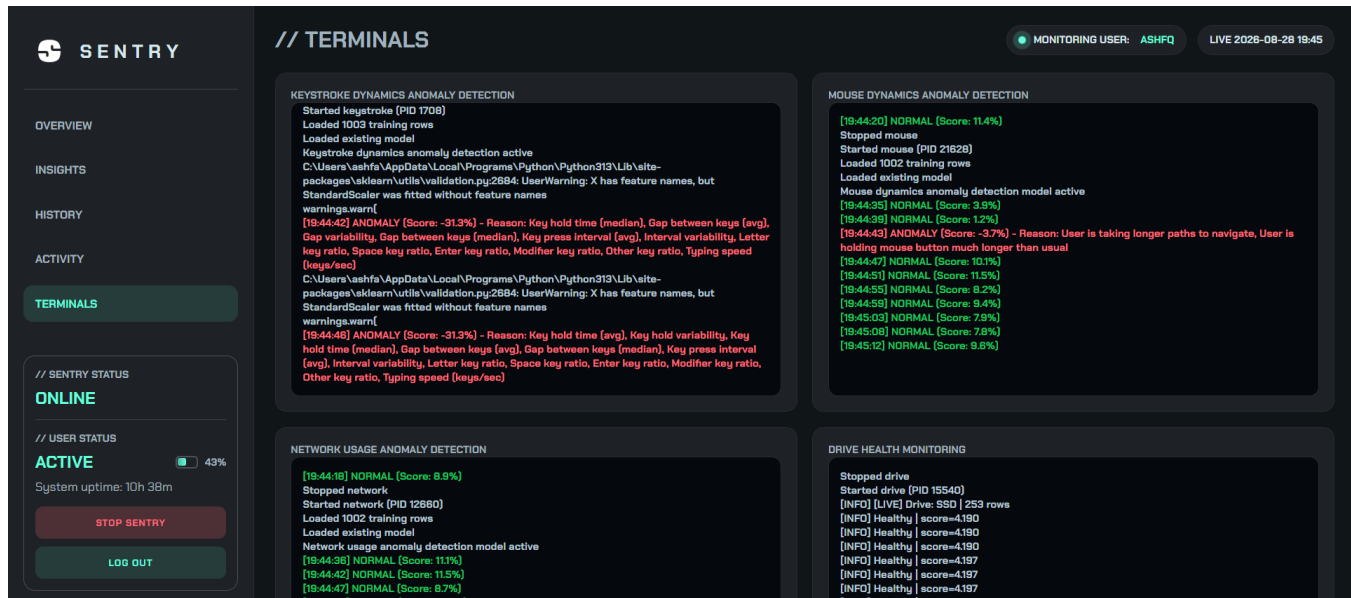
The **LOGS** page separates the Sentry activity log from the user activity log. These include component starts and stops, remote commands, and recorded activity. **DELETE LOG** permanently clears the log directories after confirmation.

Terminals and Telegram

The **TERMINALS** page shows the live output captured from running components. The component list includes the four detectors, the Telegram controller, and Telegram notifications. Use each **ENABLE** or **DISABLE** button for individual control. Telegram notifications require valid environment values and an internet connection.

An anomaly means that a detector found activity sufficiently different from its learned baseline. Review the score, detector name, timestamp, and recorded reasons before taking action. A single alert is an indication for review, not proof of an incident.

To enable Telegram notifications, make sure `.env` is configured, then turn on the Telegram setting in the dashboard. The remote bot uses the dashboard password for authentication. Remote commands may also require approval in a local confirmation window. Treat the bot token and dashboard password as sensitive credentials.



7. Remote Telegram Commands

The Telegram controller starts with Sentry when the bot token and chat ID are configured. Before using it, open the bot and send `/start`. Sentry sends an authentication prompt to authorized chats when the controller starts. Enter the dashboard password as a normal Telegram reply, or send any message and wait for the password prompt. The chat ID must be listed in `TELEGRAM_CHAT_ID`; messages from other chats are ignored and recorded as unauthorized.

After successful authentication, commands are accepted from that chat during the running controller session. With **Allow All Mode** disabled, every command displays remote command approval dialog on the monitored Windows computer. The local user must choose **Yes** before the command runs. Choosing **No**, closing the dialog, or having an approval error rejects the command. The command and decision are written to the activity log. Enable **Allow All Mode** only for a controlled period because it removes this per-command local approval step. Restarting the controller requires authentication again.

Send commands as written below. Names and command keywords are case-insensitive.

Sentry control and files

- `/stop` - stops Sentry.
- `/log` - sends the newest user activity log to Telegram.
- `/report` or `/generate report` - generates and sends the current PDF report.
- `/enable all` - starts all components.
- `/disable all` - stops all components.
- `/enable keystroke`, `/enable mouse`, `/enable network`, `/enable drive`, `/enable activity`, or `/enable remote` - starts individual components.
- `/enable telegram` - enables Telegram alerts.

- `/disable keystroke, /disable mouse, /disable network, /disable drive, /disable activity, or /disable remote` stops individual components.
- `/disable telegram` - disables Telegram alerts.
- `/delete detections` - permanently deletes anomaly detection files.
- `/delete logs` or `/delete log files` - permanently deletes the Sentry and user log directories.
- `/clearlogs` - removes older user log files while keeping the newest text log.

Applications and windows

- `/open APP` - opens an application, for example `/open notepad` or `/open chrome`.
- `/close APP` - closes matching running applications, for example `/close calculator`.
- `/getfocus` - reports the title of the active window.
- `/getallwin` - reports the titles of open windows.
- `/closefocus` - closes the active window.
- `/closeall` - attempts to close all open windows.
- `/search WORDS` - opens a Google search in the default browser, for example `/search project sentry`.

Application names supported by the built-in shortcuts include `notepad`, `calculator` or `calc`, `file explorer` or `explorer`, `command prompt` or `cmd`, `powershell`, `task manager`, `resource monitor`, `control panel`, `paint`, `snipping tool`, `character map`, `on screen keyboard`, `magnifier`, `registry editor`, `system information`, `system configuration`, `disk management`, `device manager`, `services`, `event viewer`, `windows media player` or `media player`, `edge` or `microsoft edge`, `chrome` or `google chrome`, `firefox`, `brave`, `opera`, `word` or `microsoft word`, `excel` or `microsoft excel`, `powerpoint` or `microsoft powerpoint`, `outlook`, `teams`, `vscode` or `visual studio code`, and `spotify`. Other names are treated as executable names and may fail if the application is not installed.

Notifications, sound, and display

- `/alert` - asks for a message, then displays it as a Windows notification.
- `/popup` - asks for a message, then displays it in a Windows warning dialog.
- `/speak` - asks for text, then reads it aloud through Windows speech.
- `/mute` - mutes pc audio.
- `volup NUMBER` - increases the volume by the specified number of volume-up key presses, for example `volup 3`.
- `voldown NUMBER` - decreases the volume by the specified number of volume-down key presses, for example `voldown 3`.
- `brightness NUMBER` - sets screen brightness to a percentage, for example `brightness 60`.

Power and destructive actions

- `/delete` - sends the Delete key to the computer.
- `/signout` - signs out the current Windows user.
- `/lock` - locks the Windows workstation.
- `/hibernate` - hibernates the computer.
- `/shutdown` - schedules a Windows shutdown after 30 seconds.
- `/bin` - empties the Windows Recycle Bin without an additional confirmation.

Check the exact command and arguments before approving a remote command. `/closeall`, `/delete`, `/signout`, `/shutdown`, `/bin`, and the `log/detection` deletion commands can interrupt work.

8. Logs and Generated Files

Sentry keeps its working data alongside the application:

- `detection engine/.../*_training.csv`: learned baseline samples
- `detection engine/.../*detections*.csv`: detector observations and predictions
- `sentry logs/`: Sentry event logs
- `user logs/`: activity and remote-command logs
- `reports/`: generated PDF reports
- `settings.json`: dashboard authentication and runtime settings

Do not edit training or detection CSV files while Sentry is running. Back up the CSV files and reports if they are important. Delete logs only when your retention and privacy requirements allow it.